

Miguel

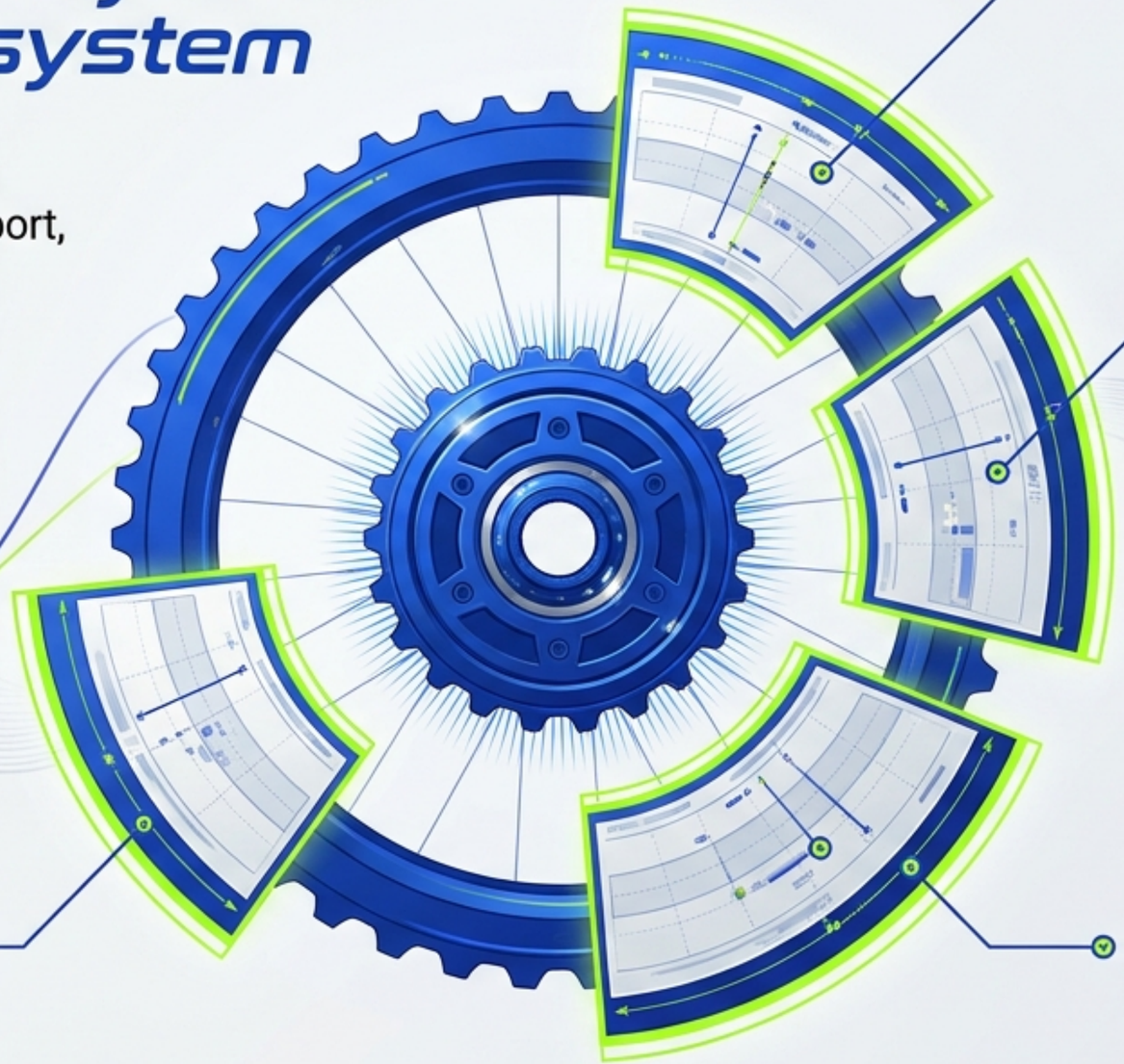
Miguel: The Intelligent Multi-Agent Companion for the Modern Cyclist.

Your ride. Your passion. My expertise.

Gemini Notebook

The Complexity of the Cycling Ecosystem

Cycling is far more than riding a bicycle. It is simultaneously a sport, a precise science, a complex mechanical pursuit, and a deeply ingrained way of life.



Professional Racing & Tactics

Race Strategy, Peloton Dynamics, Time Trialing, Climbing Efficiency.
Metrics: FTP, Power-to-Weight, Aerodynamic Drag Coefficient (CdA).

Bicycle Mechanics & Technology

Frame Material, Drivetrain Efficiency, Wheel Aerodynamics, Electronic Shifting, Suspension Systems.
Metrics: Watt Saving, Component Stiffness, Rolling Resistance.

Physiology & Training Science

VO2 Max, Lactate Threshold, Heart Rate Zones, Recovery Protocols, Periodization, Altitude Training.
Metrics: TSS (Training Stress Score), HRV (Heart Rate Variability), Recovery Time.

Nutrition & Hydration Strategies

Carbohydrate Loading, Electrolyte Balance, Intra-Race Fueling, Post-Ride Recovery, Sweat Rate Analysis.
Metrics: Kcal intake, Hydration Level, Glycogen Stores.

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Everything Cycling. Under One Roof.

Miguel is an intelligent companion built for everyone who breathes cycling—from casual weekend riders to seasoned enthusiasts and followers of the professional peloton.

- Personal Cycling Mechanic
- Race Analyst
- Nutrition Guide
- Training Companion

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The Total Addressable Expertise



Professional Racing

Grand Tours, Classics, tactics, and cycling history.



Bicycle Technology

Frames, groupsets, power meters, and emerging tech.



Training & Performance

Power zones, heart-rate training, and metrics.



Nutrition & Hydration

Fueling strategies, electrolytes, and recovery.



Bike Maintenance

Routine servicing, lubrication, and preventive care.



Repairs & Troubleshooting

Diagnosing issues, drivetrain fixes, and brake adjustments.



Cycling Equipment

Bikes, helmets, tools, and equipment comparisons.



Cycling Culture

Terminology, iconic climbs, and historic rivalries.

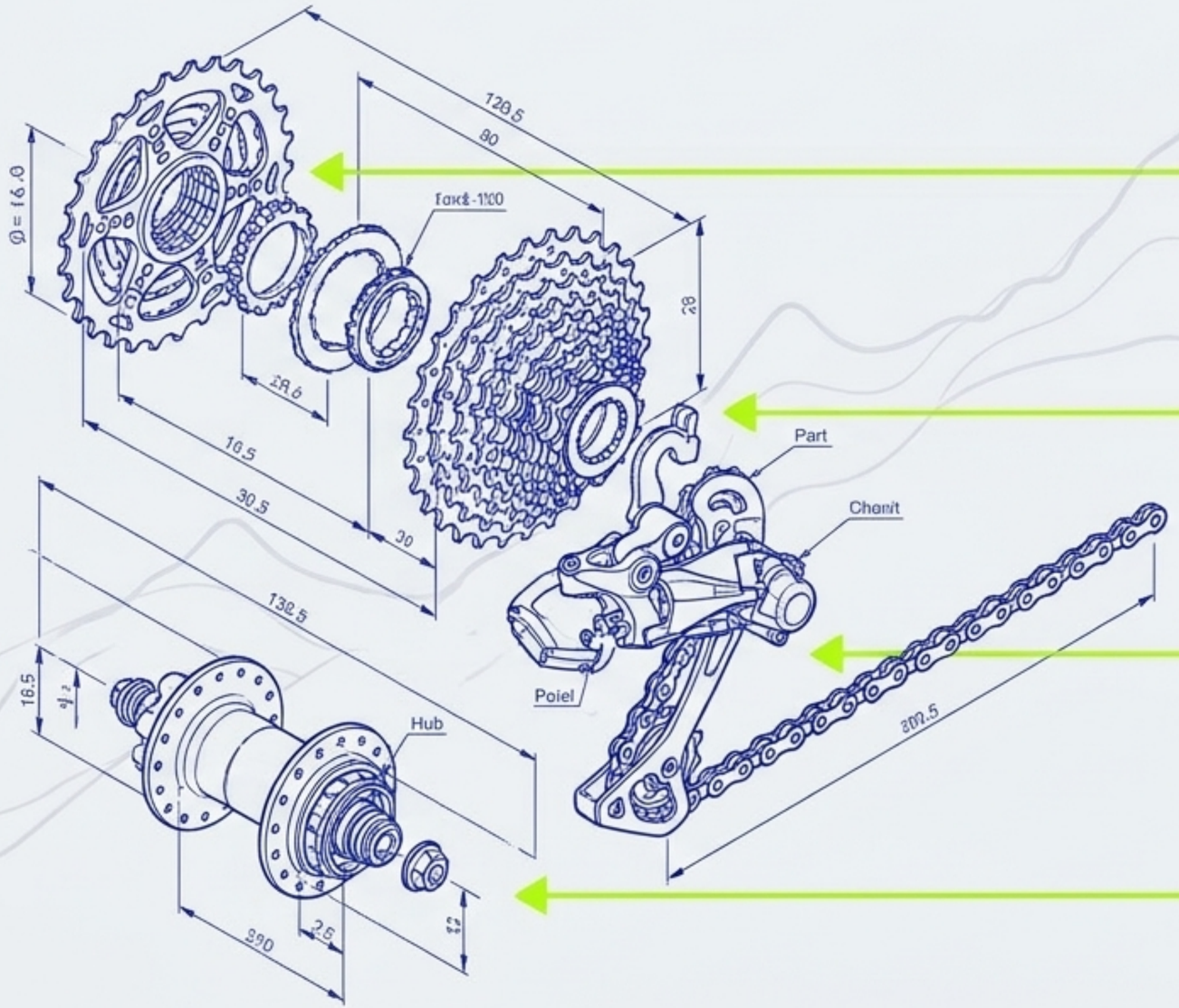
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More Than a Cycling Chatbot

	Generic AI	Miguel
Accessibility	Requires precise technical prompting.	**Conversational Intelligence:** Explore topics naturally without needing mechanical terminology. ✓
Utility	Generalized, often hallucinated mechanical summaries.	**Practical Expertise:** Actionable, step-by-step guidance for repairs and maintenance. ✓
Depth	Surface-level race results and generic Wikipedia summaries.	**Deep Cycling Knowledge:** Nuanced history, team tactics, personalities, and performance science. ✓
Memory	Isolated, episodic queries that reset every session.	**Contextual Assistance:** Builds progressively on previous conversations (remembers your bike and FTP). ✓

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The Virtual Mechanic in Your Garage



Diagnose

Troubleshoot drivetrain issues and mysterious creaks safely.

Maintain

Optimize routine servicing, cleaning, and lubrication for peak efficiency.

Repair

Step-by-step guidance for fixing punctures and executing brake adjustments without shop fees.

Upgrade

Navigate complex bicycle technology, from electronic groupsets to wheel compatibility.

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The Digital Coach & Nutritionist



Performance Metrics

Decode power zones, endurance thresholds, and heart-rate training principles.

Fueling Strategy

Precise planning for hydration, electrolytes, and race-day nutrition protocols.

Recovery Optimization

Post-ride nutrition and physiological recovery tactics.

Goal Execution

Strategic preparation for demanding climbs and century rides.

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Multi-Agent Knowledge Architecture

The Mechanic Agent

Trained on technical manuals, component specs, and repair workflows.



The Physiologist Agent



Focused on training science, power data, and biological recovery.

Miguel Context Engine

The Race Analyst Agent

Accessing Grand Tour history, stage results, and team tactics.

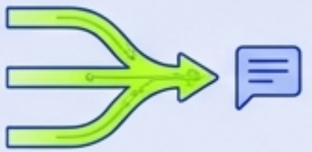


The Equipment Guru Agent



Evaluating market gear, power meters, and emerging tech.

Dynamic Routing Synthesis: When asked, 'How do I prep my bike and body for Alpe d'Huez?', the Context Engine synthesizes real-time input from the Mechanic, Physiologist, and Race Analyst simultaneously.



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Contextual Intelligence: A Companion That Learns

Interaction 1

User inputs baseline data.

Chat: I ride a mechanical Shimano 105 and my FTP is 220W.

Interaction 2 (Context Applied)

User asks a generic question.

Chat: Why is my shifting sloppy?

Miguel factors in the mechanical 105 groupset to provide specific cable-tension advice.

Interaction 3 (Deep Personalization)

User plans a ride.

Chat: What should I eat before my 100km ride today?

Miguel factors in the 220W FTP and the specific route to calculate exact carbohydrate requirements.

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Built for the Love of Cycling. Open for Everyone.

The Engine

Created and maintained by developer Tanmoy Sen (@tsen09).

The License

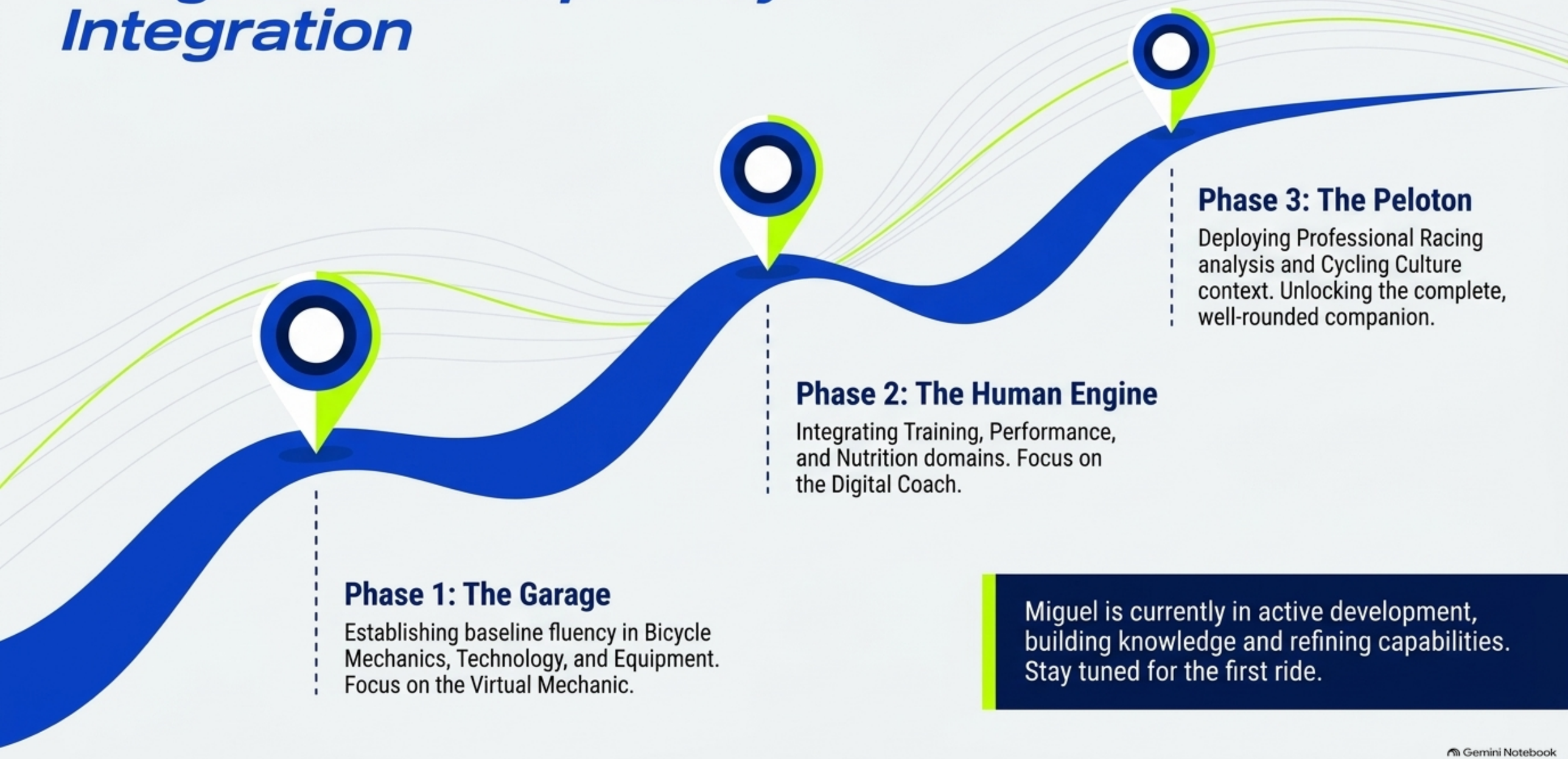
Released fully under the MIT License, enabling maximum community adoption.

The Flywheel

Open-source architecture invites contributions from the global developer and cycling communities, driving rapid expansion of knowledge sources and specialized skills.

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Progressive Capability Integration



Phase 1: The Garage

Establishing baseline fluency in Bicycle Mechanics, Technology, and Equipment. Focus on the Virtual Mechanic.

Phase 2: The Human Engine

Integrating Training, Performance, and Nutrition domains. Focus on the Digital Coach.

Phase 3: The Peloton

Deploying Professional Racing analysis and Cycling Culture context. Unlocking the complete, well-rounded companion.

Miguel is currently in active development, building knowledge and refining capabilities. Stay tuned for the first ride.

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The Holistic Cyclist Loop

Optimal Maintenance
(Frictionless mechanical performance)

Informed Inspiration
(Tactical awareness from pro racing)



Precision Training
(Actionable performance metrics)

Strategic Fueling
(Peak physical readiness)

Miguel doesn't just answer questions; he continuously optimizes the complete cycling lifestyle.

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Miguel

Every ride has a story. Every question deserves an answer.

Join the community. Build the future of sports telemetry.

GitHub: @tsen09 | MIT Licensed

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